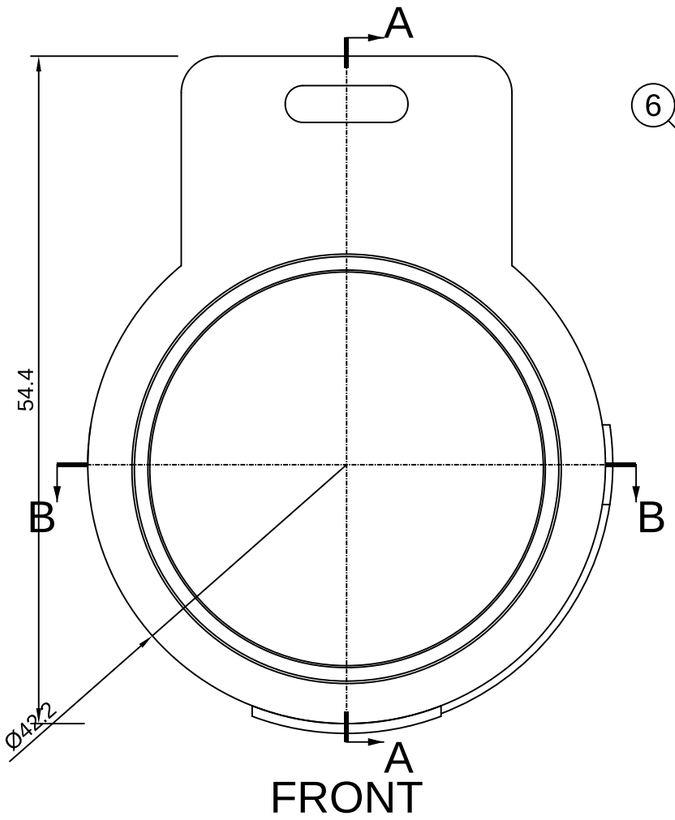
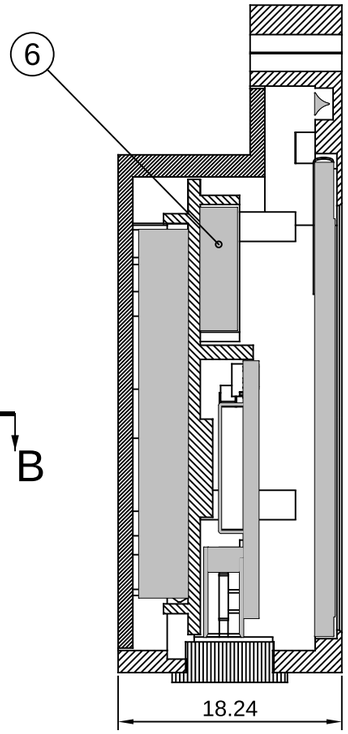
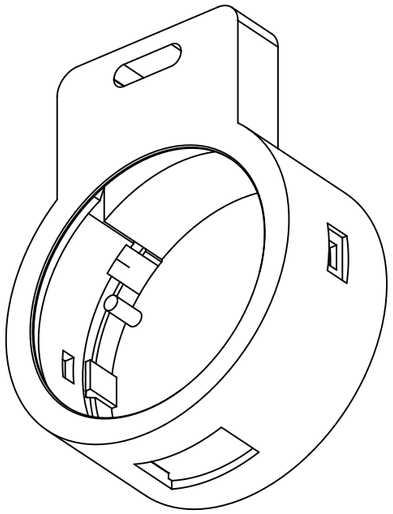


FRONT

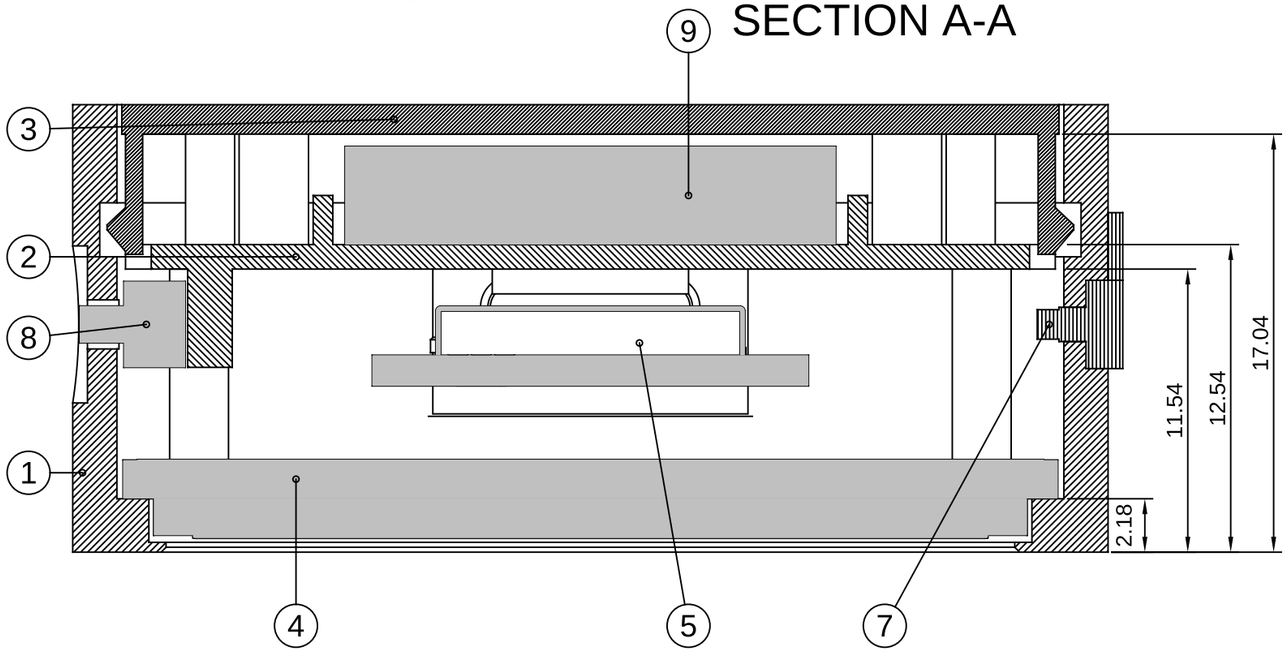


SECTION A-A



ISOMETRIC (1.4:1)

ASSEMBLY (all from the back)
1. Display (4), tab first into the lug pocket, PCB on the housing step.
2. Button (8) into its wall hole.
3. Midframe (2) with XIAO (5) and motor (6), key tabs in the 45/135 deg channels.
4. Battery (9) between the ribs; back cover (3): tab end first, then the disc.
5. USB plug (7): barb through the +X slot, then into the USB-C pocket.
6. Screen protector (10) on the clean, dry front face.
Overall ⌀42.2 x 54.4 x 18.24.
Section fill: grey = purchased parts.



SECTION B-B
(4:1)

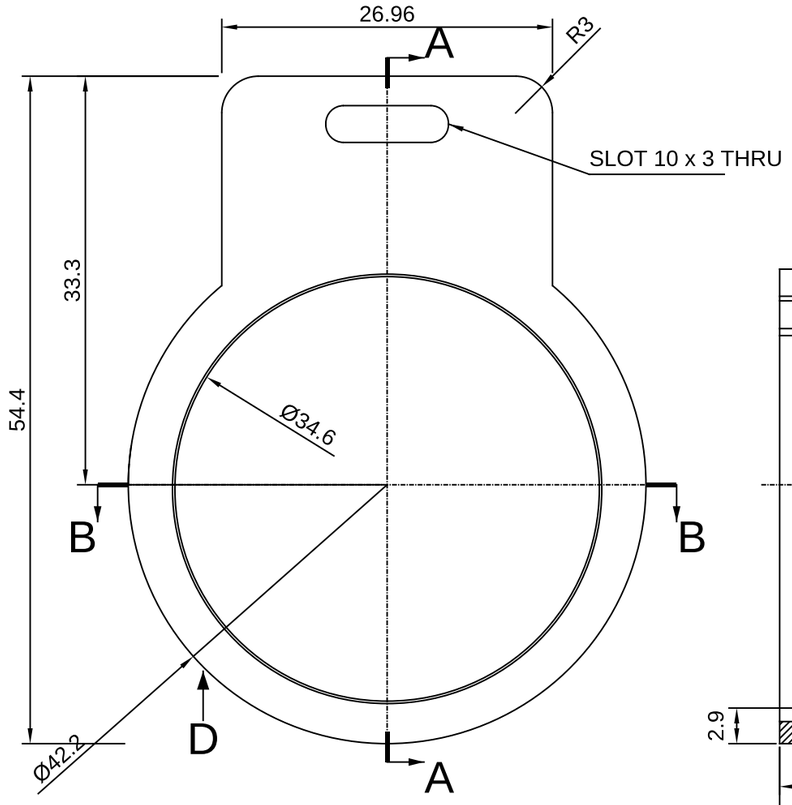
ITEM	PART	QTY	MATERIAL / NOTE
1	Front housing	1	PETG, drawing MC-M-002
2	Midframe	1	PETG, drawing MC-M-003
3	Back cover	1	PETG, drawing MC-M-004
4	GC9A01 1.28in round TFT	1	Teyleten Robot module, 7-pin header removed
5	XIAO nRF52840 Sense	1	Seeed Studio; taped to the midframe pad
6	Vibration motor 10 x 3	1	Coin type, adhesive back
7	USB-C plug, tethered	1	TPU 95A, usb_plug.step
8	Tact switch 3 x 6	1	4.3 high
9	LiPo 402030	1	3.7 V ~200 mAh, 20 x 30 x 4
10	Round screen protector	1	⌀38-40, bonded on the front face

PROJECT
MysteryChain case

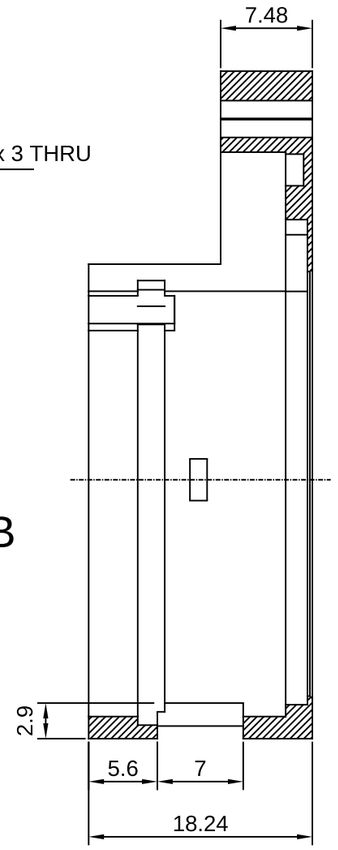
TITLE
ASSEMBLY

PROJECTION
FIRST ANGLE (ISO E)

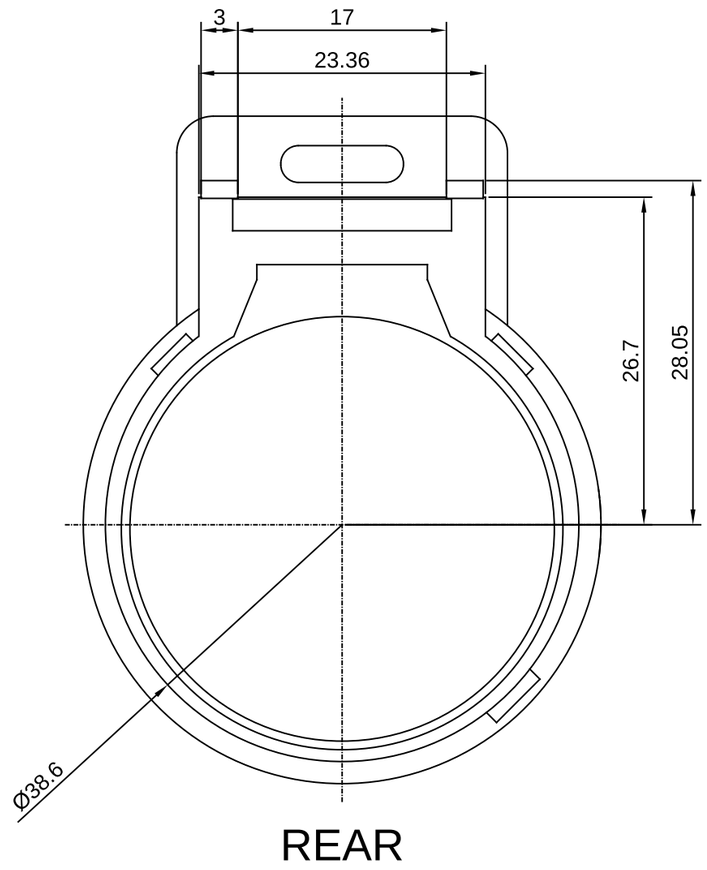
DRAWN	DATE	REVISION	DRAWING NO.
O. Eren	2026-10-15	v6	MC-M-001
MATERIAL	SCALE	UNITS / GEN. TOL.	SHEET
PETG	2:1	mm / ±0.2	1 / 4



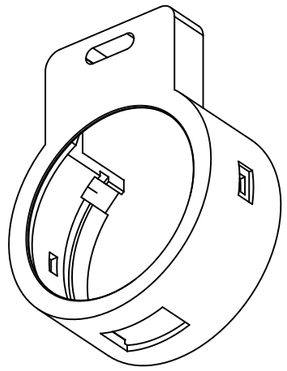
FRONT



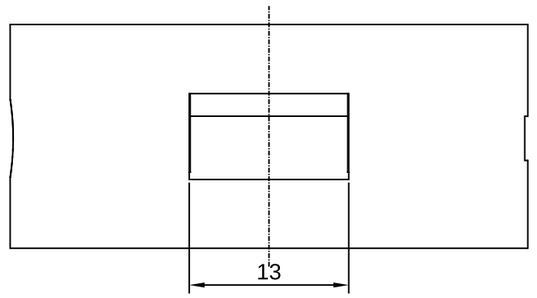
SECTION A-A



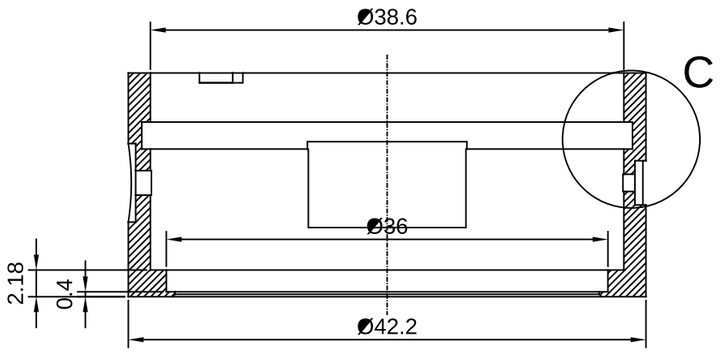
REAR



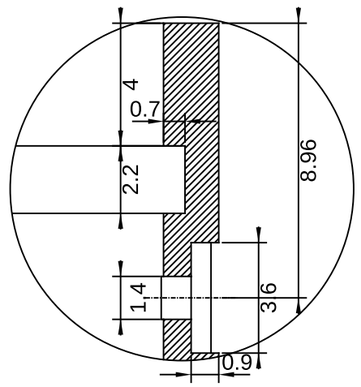
ISOMETRIC (1:1)



VIEW D



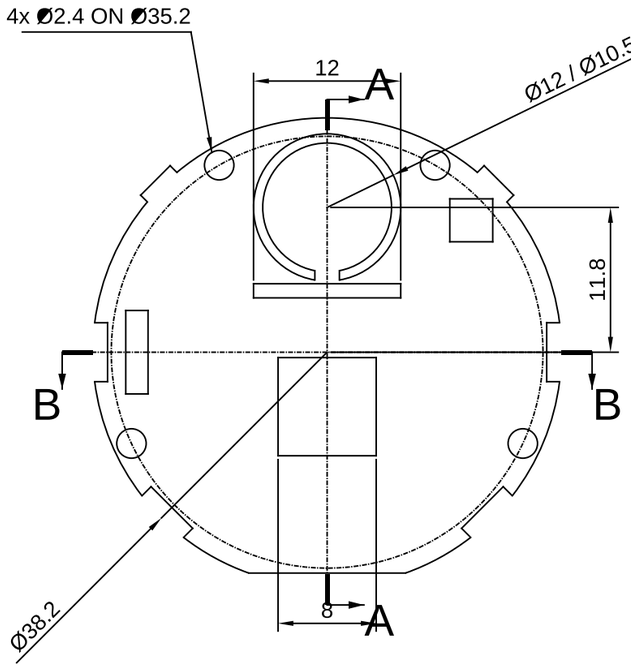
SECTION B-B



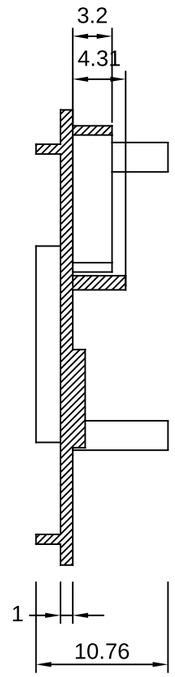
DETAIL C
(5:1)

NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Window Ø34.6 (0.4 lip over the glass edge) is centred 0.335 below the body axis (toward USB-C).
5. Key channels 2x 4 wide at 45 and 135 deg; pry notch 5 wide x 0.8 at 225 deg.
6. USB plug anchor slot 3.4 x 1.4 + thumb recess (+X, old switch opening); button hole 3.6 x 2.0 with Ø6.4 x 0.6 finger dimple (-X); both on the X axis.

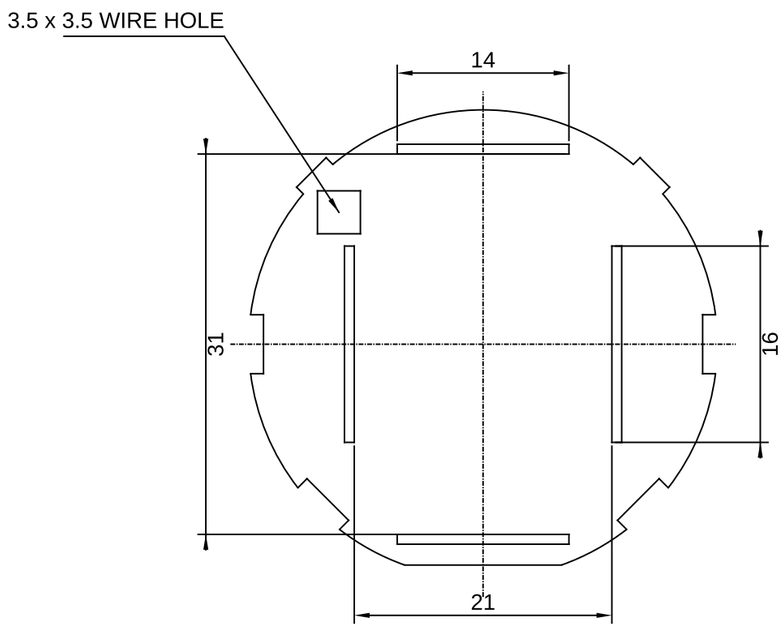
PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: FRONT HOUSING			
DRAWN: O. Eren	DATE: 2026-10-15	REVISION: v6	DRAWING NO.: MC-M-002
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 2 / 4



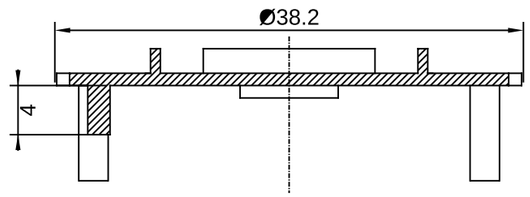
FRONT



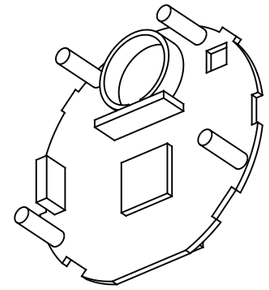
SECTION A-A



REAR



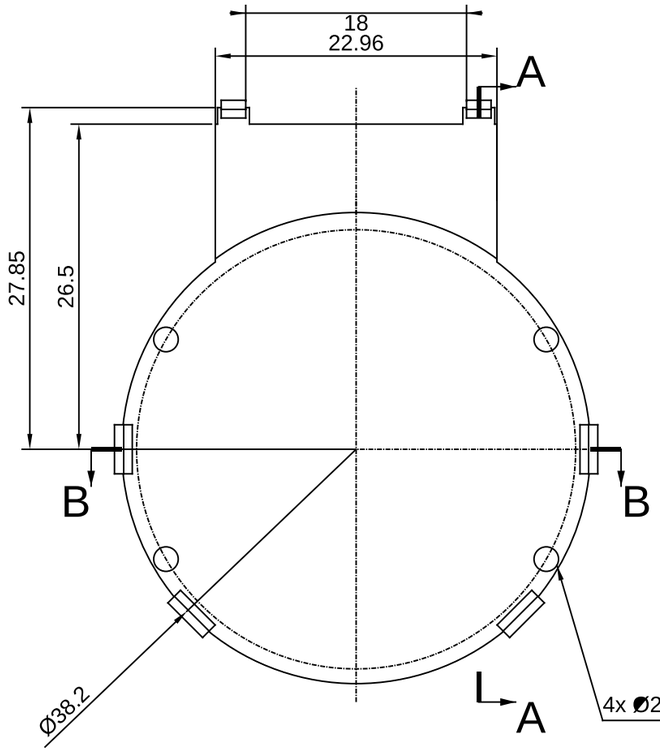
SECTION B-B



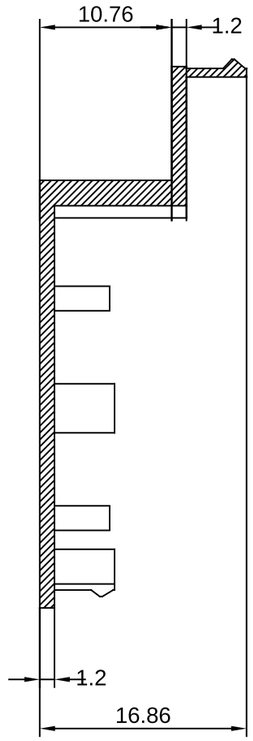
ISOMETRIC (1.2:1)

NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Display clamp legs 4x Ø2.4 at 60/120/205/335 deg press on the display PCB edge.
5. Key tabs 2x 3.4 wide at 45/135 deg (fit the housing key channels, one way only).
6. Hook notches 4x 4.8 wide to R17.9 at 0/180/225/315 deg.
7. XIAO tape pad 8 x 8 x 1.0 high for 0.5 mm double-sided tape; stop rib 12 wide at the antenna end. Battery ribs 0.8 thick x 2 high.
8. Print battery ribs down, supports touching the bed only.

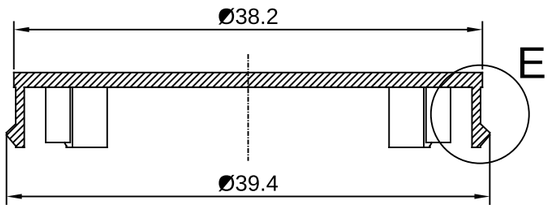
PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: MIDFRAME			
DRAWN: O. Eren	DATE: 2026-10-15	REVISION: v6	DRAWING NO.: MC-M-003
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 3 / 4



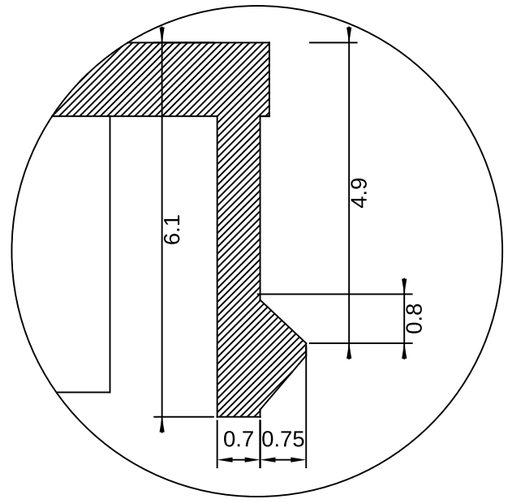
INSIDE (FROM FRONT)



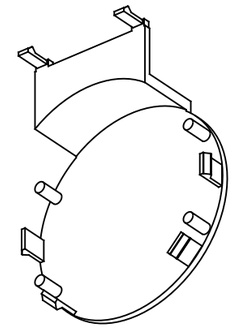
SECTION A-A



SECTION B-B



DETAIL E
(10:1)



ISOMETRIC (1:1)

NOTES
1. The 3D model pet_case.step (source pet_case.step.py) is the master; undimensioned features per model.
2. FDM, PETG, 0.4 nozzle, 0.12-0.16 layers, 3-4 walls, 30 % infill.
3. Angles are CCW from +X, seen from the front (display side).
4. Perimeter snap hooks 4x 4 wide at 0/180/225/315 deg, profile per detail E.
5. Tab-tip hooks 2x 2 wide at X = +-10, same profile, rising from the tab plate.
6. Posts 4x Ø2 at 30/150/210/330 deg clamp the midframe (top at 5.7 from the outside face).
7. Retention ramp 45 deg: the cover is removable (pry at the housing notch, 225 deg).
8. Print disc down; support under the tab plate.

PROJECT: MysteryChain case			PROJECTION: FIRST ANGLE (ISO E)
TITLE: BACK COVER			
DRAWN: O. Eren	DATE: 2026-10-15	REVISION: v6	DRAWING NO.: MC-M-004
MATERIAL: PETG	SCALE: 2:1	UNITS / GEN. TOL.: mm / ±0.2	SHEET: 4 / 4